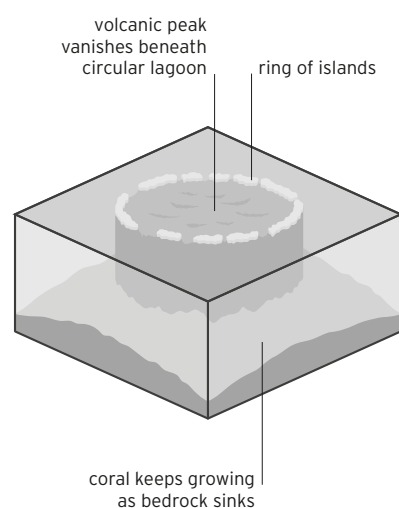


as the more gently dipping flank of the fold. See also *anticline*, *fold (panel)*, *syncline*.

ATMOSPHERE The gaseous layers that surround Earth. The atmosphere is made up of four distinct layers—the troposphere, stratosphere, mesosphere, and ionosphere. See also *stratosphere*, *troposphere*.

ATOLL ▼ A ring-shaped coral island, or a ring of small coral islands, surrounding a lagoon. Atolls are created as coral reefs build up in shallow water around subsiding volcanic islands. See also *coral reef*, *fringing reef*.



AUREOLE An area of rock around an igneous intrusion whose composition, structure, or texture has been thermally altered or metamorphosed. See also *country rock*, *intrusive igneous rock*, *metamorphism*.

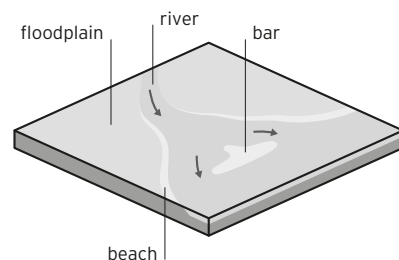
AURORA Colored streamers of light visible in the night sky at certain latitudes and at certain times of the year. They are caused by the interaction between high-energy particles from the Sun and Earth's magnetic field. In the Northern Hemisphere, displays are called the aurora borealis; the equivalent in the Southern Hemisphere is the aurora australis.

B

BACK-ARC BASIN An undersea basin formed behind an island arc near where two tectonic plates meet. See also *island arc*.

BACKSHORE An area of the shore that is above the average high-water mark. This region is covered by water only during storms and the highest tides. See also *foreshore*.

BACKWASH The flow of water down a beach after a wave has broken on the shore. See also *swash*.



BAR ▲ A linear deposit of gravel or sand deposited by a river or the sea. Bars may be exposed at low tide or permanently submerged. See also *barrier bar*, *longshore bar*, *point bar*.

BARCHAN A crescent-shaped dune formed by wind action in a sandy desert. The steeper slope, or slip face, is on the concave side. It is also called a barkhan dune or crescentic dune. See also *dune (panel)*.

BARRIER BAR A longshore bar of gravel or sand roughly parallel to the coastline, whose surface is below mean sea level. See also *barrier island*, *longshore bar*.

BARRIER ISLAND A long, thin island of sediment roughly parallel to a coastline, whose surface is usually exposed. See also *barrier bar*, *longshore bar*.

BASALT A common, fine-grained or glassy volcanic rock that usually originates from solidified lava. The oceanic crust is composed of basalt, and basaltic lava can also erupt on continents. See also *extrusive igneous rock*, *flood basalt*.

BASIN A low-lying area or depression on land or in an ocean where sediment often accumulates. See also *river basin*.

BATHOLITH A very large igneous intrusion, 60 miles (100 km) or more across, which originates deep underground. Although they form deep underground, batholiths may be exposed by erosion. See also *igneous intrusion (panel)*.

BATHYAL Referring to the zone of the ocean between about 660 and 6,600 ft (200 and 2,000 m) deep. The bathyal zone is not as deep as the abyssal zone. See also *abyssal*.

BAYMOUTH BAR A spit or bar that extends completely across the mouth of a bay. See also *barrier bar*, *spit*.

BEACH See panel, below.

BEACH FACE The sloping section of a beach, which is exposed to the wash of the waves. Also called the foreshore. See also *beach (panel)*, *swash*.

BEDDING The sequence in which layers of sedimentary rock were originally laid down, one on top of another. See also *bedding plane*, *sedimentary rock*.

BEACH

A beach is an accumulation of sand, gravel, shell debris, or other deposits along the shoreline of an ocean, lake, or river. Ocean beaches are constantly changing as tides and wave action deposit and carry away sediment. Every beach has a distinct profile above and below the water line. The intertidal zone lies between the low- and high-water marks.

